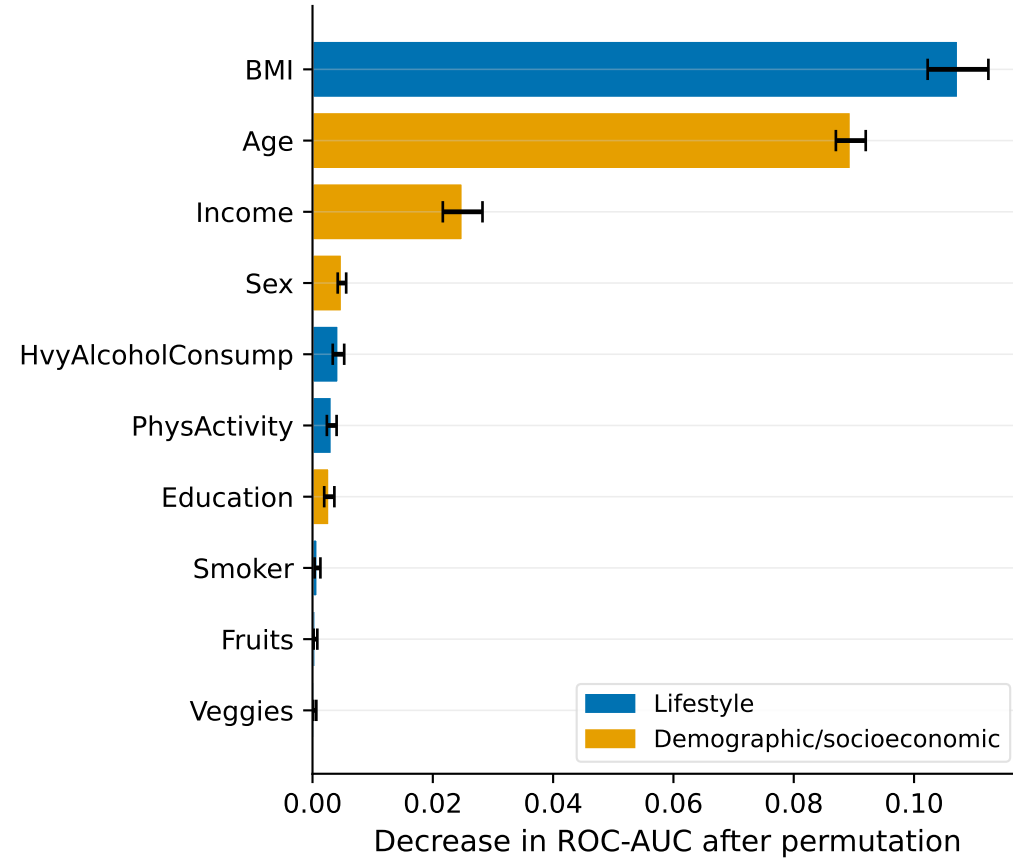
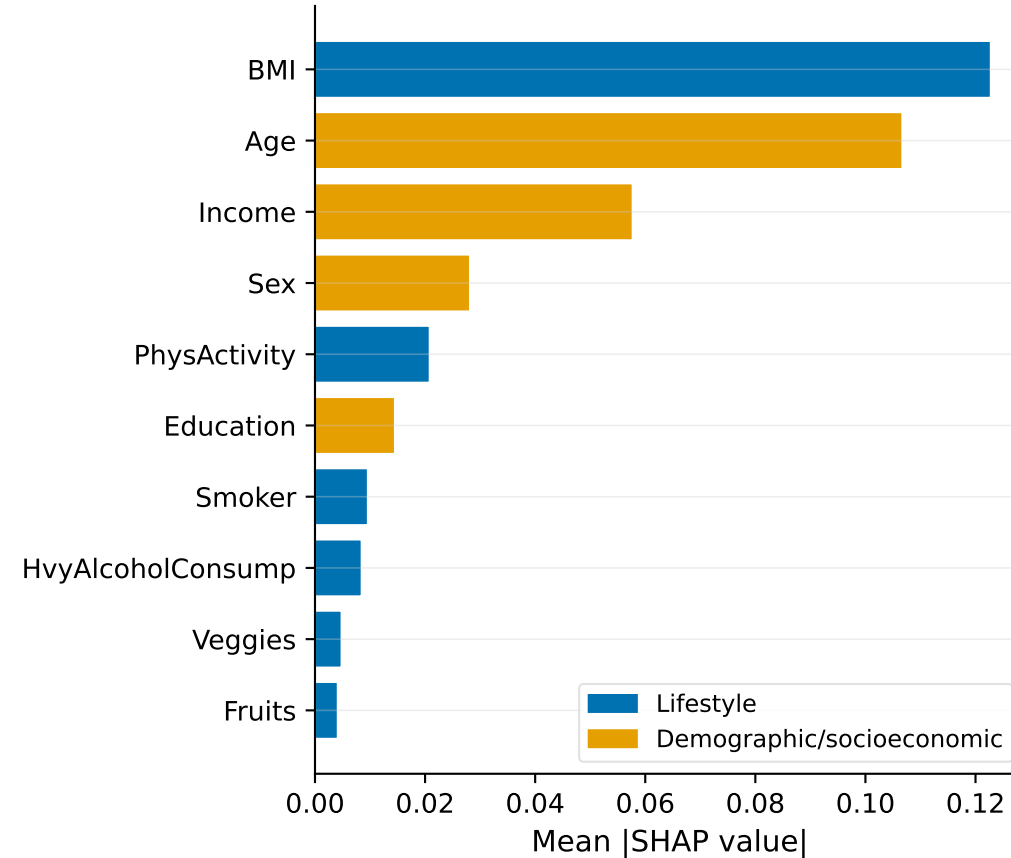


RQ2: Feature contribution and interpretability

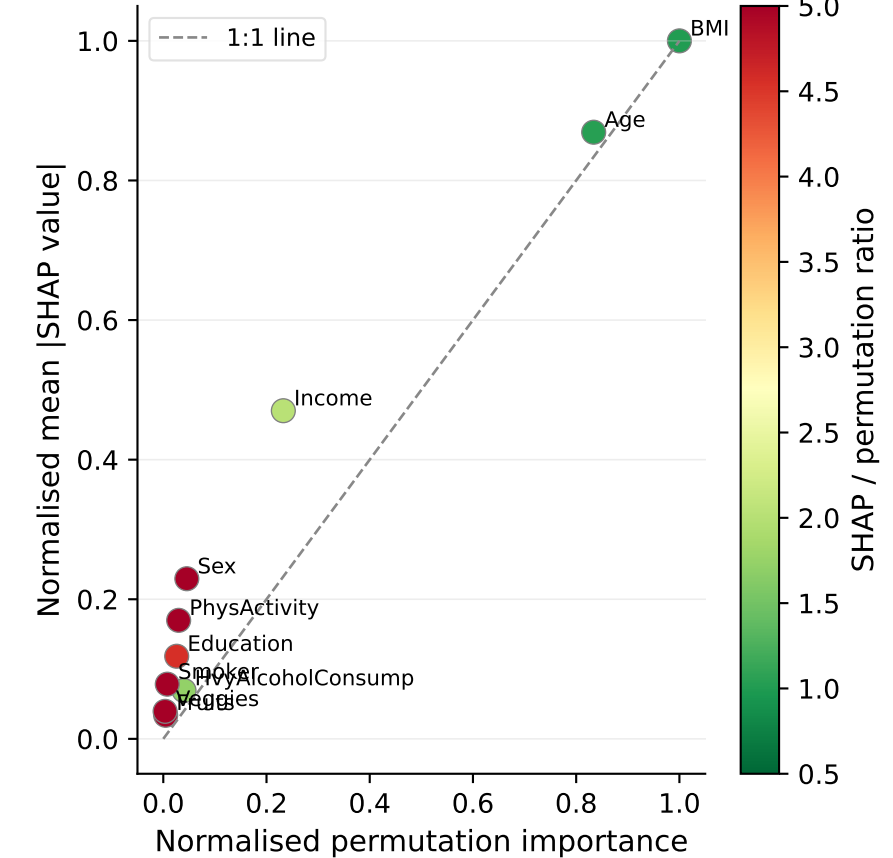
A (A) Permutation importance



B (B) SHAP feature importance



(C) Permutation vs SHAP consistency
(colour = SHAP/permutation ratio; warm = SHAP > permutation)



Note (M3): Lifestyle features (Fruits, Veggies, Smoker, PhysActivity, HvyAlcohol) show higher SHAP than permutation importance, consistent with known permutation-importance underestimation in correlated feature clusters. Permutation importance is reported for verification; SHAP is the primary interpretability metric.